

Lesson 08 Practice: Answers and Rationales

JKSSB Computer Awareness

October 10, 2026

Provenance: All 20 items are *[Original]* JKSSB-pattern questions for this lesson. No item is a real PYQ, and the answer key is the author's verification of the lesson's concepts.

- Q1. C.** RAM is volatile, so it loses its contents when power is removed. ROM, a hard disk, and a flash drive are non-volatile and retain their contents.
- Q2. A.** RAM stands for Random Access Memory. B, C, and D are invented expansions.
- Q3. A.** ROM stands for Read-Only Memory. B, C, and D are invented expansions.
- Q4. C.** RAM is volatile and does *not* retain its contents after power-off, so C is the statement that is NOT correct. A, B, and D are all true statements.
- Q5. B.** RAM is the volatile main memory holding running programs and their data. ROM is non-volatile; L1 cache is smaller working memory; the SSD is secondary storage.
- Q6. A.** DRAM is a volatile form of RAM. ROM, PROM, and EPROM belong to the non-volatile ROM family.
- Q7. D.** SRAM is faster than DRAM and needs no refresh, which is why it is used for cache. DRAM needs periodic refresh; ROM and a hard disk are non-volatile and slower than SRAM.
- Q8. D.** DRAM is denser and cheaper than SRAM but needs periodic refresh. A describes SRAM; B is the use of SRAM; C is false, since DRAM is volatile.
- Q9. B.** SRAM is used for cache memory because it is fast and needs no refresh. A is DRAM's role; C and D describe other memory classes.
- Q10. A.** DRAM serves as main memory: dense and cheap, at the cost of periodic refresh. B is SRAM's role; C describes ROM; D is a register, not DRAM.
- Q11. B.** EEPROM is Electrically Erasable Programmable Read-Only Memory. A should be Dynamic, C should be Erasable (UV), and D should be Static, so each of the others is a wrong expansion.
- Q12. D.** PROM is programmable once and cannot be erased. EPROM and EEPROM can be erased; DRAM is a volatile RAM, not a ROM type.
- Q13. B.** EPROM is erased by ultraviolet light. PROM cannot be erased, EEPROM is erased electrically, and mask ROM is fixed at manufacture.
- Q14. C.** EEPROM is erased and rewritten electrically, in-circuit. PROM cannot be erased, EPROM needs UV light plus removal, and SRAM is volatile RAM.

- Q15. B.** Flash is an EEPROM-derived non-volatile memory erased in blocks. A wrongly makes it volatile DRAM; C describes a fixed ROM type; D is a register, not flash.
- Q16. A.** Statements I and II are correct: RAM is volatile and ROM is non-volatile, and ROM is normally read-only though specially programmable. III is false because cache is faster than main memory, not slower. B and C each include false III; D includes all three.
- Q17. A.** L1 is the smallest, fastest cache and sits closest to the core. L2 and L3 are larger and slower; main memory is slower still.
- Q18. D.** L3 is the largest cache level and is typically shared across cores. L1 and L2 are smaller and faster; ROM is not cache.
- Q19. C.** A miss means the data is not found in the cache, so it is fetched from main memory. A describes a hit; B and D are unrelated.
- Q20. D.** Hit ratio = hits $\div$ total accesses = $90/100 = 0.90$. A is the miss fraction, B would mean every access hit, and C is arithmetically wrong.
- Q21. A.** RAM is volatile and readable/writable. B, C, and D wrongly make it permanent or tie it to peripherals.
- Q22. A.** Firmware lives in non-volatile ROM/Flash. RAM (B) loses contents; C and D are not firmware stores.
- Q23. A.** SRAM's speed suits cache. Tape (B), floppy coating (C), and paper (D) are not cache technologies.
- Q24. A, B, C.** RAM, cache, and registers lose contents without power. Masked ROM (D) is non-volatile.
- Q25. A, B, C.** DRAM refresh, SRAM speed, and SRAM-based cache are correct. D is false: ROM keeps contents without power.
- Q26. B.** RAM is volatile, so its contents are lost when power is removed. It is not automatically copied to ROM or cache.
- Q27. B.** Firmware must be available at startup and stored in non-volatile memory, traditionally ROM. The other listed memories are not the standard firmware store.
- Q28. A.** RAM loses its contents without power. ROM, SSDs, and optical discs are non-volatile.
- Q29. B.** SRAM uses flip-flop cells and does not require periodic refresh. DRAM uses capacitors and must be refreshed; ROM and flash are non-volatile.
- Q30. A, B, C.** RAM is normally volatile, ROM retains data, and DRAM needs refresh. SRAM is generally faster than DRAM, not slower.